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Finding the Signal in the Spam: Jointly Learning Rewards and Worker Reliability from Pairwise Comparisons

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Introduction

- Reward learning from pairwise comparisons is a widely studied problem in machine learning.
- Given pairwise comparisons between K items, the goal is to learn a reward for each item that is *"consistent"* with the observed comparisons. However, crowdworkers often exhibit large variability in the quality of labels.
- To address this, we adopt the ***Boltzmann-rational model,*** in which each item $j \in [K]$ has a latent reward r_j and each worker $s \in [M]$ and $\beta_s \in [-1, 1]$ as their competency parameter. The probability that worker s prefers item w over item l is given as: $P[w \succ l \mid s] = \sigma(\beta_s(r_w - r_l))$, where σ is the logistic function.

Goal: Jointly Estimate Item rewards (**r**) and Worker competencies (**β**) from pairwise comparison data.

Prior Work

- **Bradley-Terry-Luce (BTL) model:** $P[w \succ l] = \sigma(r_w - r_l)$
- **Rank Centrality (RC):** interprets pairwise comparisons as a directed graph, where nodes represent items and directed edges represent comparison outcomes, using iterative aggregation techniques, gets the reward values.
- **CrowdBT:** $\Pr(w \succ l \mid s) = \beta_s\, \sigma(r_w - r_l) + (1 - \beta_s)\, \sigma(r_l - r_w)$
- **FactorBT:** $\Pr(w \succ l|s) = \sigma(\gamma_s) \cdot \sigma(r_w - r_l) + (1 - \sigma(\gamma_s)) \cdot \sigma(\langle \mathbf{b}_s, \mathbf{x}_{swl} \rangle)$
  - $\sigma(\gamma_s)$ is similar to annotator competence, $\boldsymbol{b}_s$ encoder worker specific sensitivity to task features and $\boldsymbol{x}_{swl}$ is a feature vector describing a comparison.
- **HBTL and HTCV:** HBTL models the Boltzmann Rational Model whereas HTCV uses the formulation as $P(w \succ l; s) = \Phi\left(\frac{\beta_s(r_w - r_l)}{\sqrt{2}}\right)$. Both use alternating gradient descent to obtain the estimates.
- **BARP:** $P(w \succ l|s) = \sigma((r_w - r_l) + \theta_s(\delta_{wg} - \delta_{lg}))$ where δ_{wg} represents the bias of evaluator s for the group g of item w.
  - *BTL, RC and BARP do not model worker competence.*
  - *No method for Boltzmann Rational Model with closed-form updates.*

Notation

K items, M workers, N comparisons $\eta_{sij} = \beta_s \sigma(r_i - r_j)$

$$\boldsymbol{\theta} = (\{\beta_s\}_{s=1}^{M}, \{\boldsymbol{r}_i\}_{i=1}^{K}) \qquad \eta_{sij} = \beta_s \sigma(r_i - r_j)$$

Note: The Boltzmann Rational Model is invariant to additive scaling in r, hence we impose the centering constraint $\sum_i r_i = 0$ to ensure identifiability. Also, it is invariant to scale transformation in r and β *e.g.,* r → cr, $\beta_s \to \beta_s/c$, we normalize rewards to unit root-mean-square to tackle this.

BoRaEM Algorithm

$$\mathcal{L}_{\text{complete}}(\boldsymbol{\theta}; \mathcal{D}) = \prod_{(w_i, l_i, s_i) \in \mathcal{D}} \sigma\left(\beta_{s_i}(r_{w_i} - r_{l_i})\right)$$

Observed-data likelihood under the Boltzmann-Rational model

↓

$$\frac{1}{1 + e^{-\eta}} = \frac{1}{2} e^{\eta/2} \int_0^\infty e^{-\omega\eta^2/2}\, p(\omega)\, d\omega$$

Identity from [1]: Introducing a latent Polya-Gamma (PG) random variable ω for each observation turns the logistic form into Gaussian

↓

$$p(w_i, l_i, s_i, \omega_i | \boldsymbol{\theta}) \propto \exp\left(-\frac{\omega_i \eta_i^2}{2} + \frac{\eta_i}{2}\right) p(\omega_i)$$

Augmented joint density

Initialize $\theta^{(0)}$ with rewards as zeros and betas with any constant value (say 0.8)

E-step

PG distribution moments lead to a *closed form* expectation over ω:

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = \sum_{(w_i, l_i, s_i) \in \mathcal{D}} \left[\tfrac{1}{2}\, \beta_{s_i}\, (r_{w_i} - r_{l_i}) - \tfrac{1}{2}\, \kappa_i^{(t)}\, \beta_{s_i}^2\, (r_{w_i} - r_{l_i})^2\right]$$

where $\kappa_i^{(t)} = \frac{1}{2\, \eta_i^{(t)}} \tanh\left(\frac{\eta_i^{(t)}}{2}\right)$, if $\eta_i^{(t)} \neq 0$ and $\kappa_i^{(t)} = \frac{1}{4}$, if $\eta_i^{(t)} = 0$.

M-step

In order to maximize the $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)})$ function, we use the Alternating Maximization (AM) algorithm. Let $d_i = r_{w_i} - r_{l_i}$ denote the reward difference between the preferred item w_i and the losing item l_i, evaluated at the current item reward estimates within the inner iterations of the M-step.

Updating Competencies $\beta_s = \frac{\sum_{i \in I_s} d_i}{2 \sum_{i \in I_s} \kappa_i^{(t)} d_i^2}$, where $I_s = \{i \in [N] : s_i = s\}$

Updating Rewards is equivalent to solving a linear system: $H\mathbf{r} = \mathbf{b}$

Under centering constraint, H is positive definite; thus unique solution.

Repeat until convergence inside the M-step.

Repeat until convergence for the entire EM Algorithm.

Theoretical Analysis

M-step objective can be written as a rank-1 matrix sensing problem: $\min_{\beta, r} \|\mathcal{A}(\boldsymbol{\beta}\boldsymbol{r}^T) - \mathbf{u}\|_2^2$, where $\mathbf{u} \in \mathbb{R}^N$ and the linear operator $\mathcal{A}: \mathbb{R}^{M \times K} \to \mathbb{R}^N$ as $\mathcal{A}(X) = (\langle A_1, X \rangle, \ldots, \langle A_N, X \rangle)^\top$ with matrices $A_i = 2c_i\alpha\, \boldsymbol{e}_{s_i}(\boldsymbol{e}_{w_i} - \boldsymbol{e}_{l_i})^\top$ where $\alpha = \sqrt{MK/2N}$, $c_i^2 = \kappa_i^{(t)}$ for iteration t, e_x denotes a unit vector having value as 1 at index x.

Sensing operator $\mathcal{A}$ satisfies the Restricted Isometry Property (RIP) with high probability, given enough comparisons N.

Under RIP, w.h.p. every local minimum of the M-step is close to the global optimum. Distance to optimum is controlled by the rank-1 approximation error [2]

$$\text{DIST}(\boldsymbol{\theta}^{(t+1)}, (\boldsymbol{\theta}^*)^{(t+1)}) \leq \frac{1250}{3\sigma_1(X_t^*)} \cdot \|\mathcal{A}(X_t^* - X_{1,t}^*)\|_2$$

Combined with monotonic AM updates, the EM likelihood increases every iteration and converges to a stationary point.

Experimental Results – Real Data

Table 1. Benchmark results on FaceAge and Passage datasets. Accuracy and Weighted Accuracy are reported as percentages, while Kendall's Tau is reported as a decimal. Best overall results are shown in **bold**.

Method	Accuracy (%)	Weighted Accuracy (%)	Kendall's Tau	Runtime (s)
FaceAge Dataset				
Simple BT	79.01	87.20	0.5754	0.89 ± 0.05
RC	78.04	86.44	0.5582	0.07 ± 0.03
FactorBT	79.16	87.30	0.5785	169.41 ± 1.16
BARP	79.08	87.26	0.5769	51.08 ± 2.85
CrowdBT	79.17	87.31	0.5787	138.16 ± 3.86
HTCV	79.17	87.33	0.5786	41.31 ± 1.41
HBTL	79.20	87.35	0.5793	83.10 ± 3.32
BoRaEM (Ours)	**79.21**	**87.36**	**0.5795**	6.55 ± 0.43
Passage Dataset				
Simple BT	68.04	74.33	0.3407	1.39 ± 0.13
RC	65.22	71.30	0.2892	0.02 ± 0.04
FactorBT	69.47	75.58	0.3676	100.85 ± 2.82
BARP	68.19	74.51	0.3435	32.05 ± 2.38
CrowdBT	**70.02**	**75.96**	**0.3779**	84.25 ± 1.84
HTCV	69.30	75.30	0.3643	46.19 ± 1.83
HBTL	69.53	75.46	0.3688	77.72 ± 2.85
BoRaEM (Ours)	69.59	75.56	0.3698	8.28 ± 0.71

References

[1] Polson, N. G., Scott, J. G., and Windle, J. Bayesian inference for logistic models using p ´olya–gamma latent variables. Journal of the American Statistical Association, 108(504):1339–1349, 2013. doi: 10.1080/01621459.2013.829001.

[2] Dohyung Park, Anastasios Kyrillidis, Constantine Carmanis, and Sujay Sanghavi. Non-square matrix sensing without spurious local minima via the Burer-Monteiro approach. In Aarti Singh and Jerry Zhu, editors, Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, volume 54 of Proceedings of Machine Learning Research, pages 65–74. PMLR, 20–22 Apr 2017. URL https://proceedings.mlr.press/v54/park17a.html